

Moses (Eli) Katz

david@smartcomputerinc.com | 301-792-9537 | College Park, MD

Website: <https://0edk.github.io/> | LinkedIn: mores-d-katz

Education

University of Maryland

B.S., Electrical Engineering, GPA 3.86

College Park, MD

Expected May 2028

Skills

Programming: Python, C, JavaScript, Go, Prolog, Lua, Haskell, Zig

Software: Linux command line, shell scripting, Verilog, HTML, CSS, Markdown, Microsoft Excel, L^AT_EX, Typst, ffmpeg, ImageMagick

Laboratory: oscilloscope, woodshop tools, electric circuit analysis

Language: Spanish, Hebrew

Independent Projects

Anki add-ons

Aug. 2025-Present

- Integrate Python, markup, and DSLs to augment existing complex software
- Apply graph theory and other pure maths for novel studying strategies

Zigzag esolang compiler

Aug. 2026

- Compiled a bespoke language to QBE's intermediate representation with a C program

Capacitive keyboard

July 2026-Aug. 2026

- Interfaced an Arduino to a GT911 capacitive touch panel via I²C
- Applied Voronoi diagrams to map touch events to keypresses
- Implemented OS-level keyboard behaviour with Linux's userland input device

Functle

July 2025-Feb. 2026

- Devised and implemented JavaScript/CSS maths puzzle game

DKL9

June 2023-Feb. 2026

- Published 150+ posts to document activities and concepts

GULiVer graph syntax

Aug. 2025-Oct. 2025

- Designed terse and flexible notation for arbitrary directed and undirected graphs
- Prototyped parser to compile custom format to Graphviz and TGF
- Enhanced editing of custom format with Vim script

Roraer input method

July 2025-Oct. 2025

- Tailored a keyboard layout to work efficiently on the structure of source code
- Introduced fuzzy-matching features to reduce keystrokes by 37%
- Enabled real-time text editor enhancements with the systems-level performance of Zig

Diverse-Order Sampling by Trie

July 2025

- Noticed and addressed inefficiency in Linux package search results
- Iterated candidate algorithms to balance performance and quality
- Applied the trie data structure in Python to reorder lists of strings

voxels

June 2025-July 2025

- Developed a 3D game to interact with ~100 complex objects
- Optimised program to run at 30+ FPS despite limits of Python and old hardware

Memoire study software

Nov. 2021-May 2025

- Realised cognitive science's theory of spaced repetition in Lua programming
- Serialised software flashcards to load 14000+ studiable facts in <500 ms

Zcourse

Dec. 2024-Apr. 2025

- Programmed a novel Arduino-based device to intelligently sense and influence sleep
- Collected day-long sensor logs under kilobyte-scale memory constraints

Lifestyle observational study

July 2024-Feb 2025

- Recorded 10 variables daily over months in Excel
- Analysed effects with correlation coefficients and conditional distributions
- Performed randomised controlled trial on effects of music

Desmos demos

Sept. 2020-May 2024

- Demonstrated quantum electrodynamics, 3D geometry, physical chemistry, and more
- Built 600+ graphs, many interactive
- Pushed limits of a barely-Turing-complete atypical programming environment
- Implemented the board game Go

Collaborative Work

University of Maryland Department of ECE

Undergraduate Teaching Fellow

College Park, MD

Aug. 2026-Present

- Compose problems to practise digital logic course material
- Guide discussion sections of 10+ students

Local K.L.T.

Organising Servant

Kemp Mill, MD

July 2026-Present

- Organised diverse books and files to fill 80+ boxes
- Took down and reconfigured computer hardware
- Guided effective purchases of video hardware and internet service

The Go Club at UMD

President

College Park, MD

Sept. 2024-May 2026

- Organised 8 meetings per month with 10+ participants

University of Maryland Over Terrain Vehicle project

Team Member

College Park, MD

Sept. 2025-Dec. 2025

- Designed and implemented wiring of robot with 4-wheel drive and 4 sensors
- Pinpointed limitations of flawed hardware to maintain progress on mission
- Contributed to 3 design presentations with Google Slides and CAD

FIRE Quantum Machine Learning

Team Member

College Park, MD

Jan. 2025-Sept. 2025

- Replicated and taught contemporary quantum computing research with Qiskit